A Single Droplet-Printed Double-Side Universal Soft Electronic Platform for Highly Integrated Stretchable Hybrid Electronics

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Soft features in electronic devices have provided an opportunity of gleaning a wide spectrum of intimate biosignals. Lack of data processing tools in a soft form, however, proclaims the need of bulky wires or low-performance near-field communication externally linked to a "rigid" processor board, thus tarnishing the true meaning of "soft" electronics. Furthermore, although of rising interest in stretchable hybrid electronics, lack of consideration in multilayer, miniaturized design and system-level data computing limits their practical use. The results presented here form the basis of fully printable, system-level soft electronics for practical data processing and computing with advanced capabilities of universal circuit design and multilayer device integration into a single platform. Single droplet printing-based integration of rigid islands and core–shell vertical interconnect access (via) into a common soft matrix with a symmetric arrangement leads to a double-side universal soft electronic platform that features site-selective, simultaneous double-side strain isolation, and vertical interconnection, respectively. Systematic studies of island-morphology engineering, surface-strain mapping, and electrical analysis of the platform propose optimized designs. Commensurate with the universal layout, a complete example of double-side integrated, stretchable 1 MHz binary decoders comprised of 36 logic gates interacting with 9 vias is demonstrated by printing-based, double-side electronic functionalization.

1. Introduction

Introducing softness in electronic devices and circuits has substantially advanced a new class of electronics that can interact with humans without any mechanical and electrical failure;[1–6] as well as contributed to developments in novel materials[7–11] and architectural modeling.[12–14] Nowadays, on the basis of accumulated knowledge, researchers have agreed that a system-level integration of soft electronics beyond developing a single device, sensor, or electrode design[15–17] is a promising aim. To accomplish an entirely stretchable system, the following four aspects should be fulfilled in a common substrate: stretchable forms of i) independent power sources (batteries or wireless power transmission circuits), ii) receptors (transducers or sensor networks), iii) data processing and computing circuits, and iv) displays, with form factors that prevent active devices from multidirectional mechanical stresses. Numerous efforts have been made to suggest a promising pathway with regard to materials and device architectures for the key technologies of power sources (i), sensors (ii), and displays (iv). Representative examples include textile concepts of stretchable batteries,[18–20] stretchable antennas,[21,22] nanostructure soft sensors,[23–25] and intrinsically stretchable forms of display.[26,27] In case of implementing stretchable data processing circuits (iii), however, a very few limited range of studies has been reported so far. Difficulties in integrated circuit (IC) design and fabrication steps directly onto a soft substrate have impeded further studies on addressing the capability of "in-system" data processing and computations. This consequently declares the need of "rigid" circuits (printed circuit boards (PCBs))–mostly connected with bulky wires–as computational backbones in most soft electronic systems.

In recent years, as an alternative to the foregoing limitations, a hybrid concept of flexible, stretchable electronics that brings together soft and hard electronics into a single platform has been arising.[28–31] In this concept, soft materials and designs construct stretchable interconnects and conformal interfaces; and cointegrated rigid silicon-based devices and circuits enable high-speed data processing and computation. Hence, both of high level system integration and advanced computational functionality are simultaneously achieved. A few strategies for reliable integration of rigid devices into a soft platform have been introduced to alleviate the dissimilarity with conventional PCB technology: microfluidic assembly,[29] transfer-combined modified flip-chip bonding,[30] and reflow soldering on a flexible substrate.[31] In most approaches, however, complexity in fabrication process (multistep photolithography) and limited design freedom (several photo-/shadow-masks) restricted the ability to form advanced networks between numbers of IC devices, concurrently lacking

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in-depth consideration of multilayer circuit designs for high level integration and operating conditions (speed, delay, etc.). Moreover, general use of bulky size chips (>4–5 mm, one side) degraded system-level stretchability and skin-conformability. Central to these challenges is the proposed use of our previous approach:[32] multistep inkjet-printed rigid islands embedded in a soft substrate effectively supported IC devices, and image-based circuit design showed potential for implementing versatile networks between several types of devices. However, multidrop, multistep printing process for island formation could be a hurdle toward rapid, large-area implementation. Also, absence of methodology for integrating stretchable vertical interconnect access (via) into a common soft substrate as well as of additional consideration in island morphology obstructed multilayer circuit designs. Although a novel concept of stretchable core–shell via was recently developed,[33] there still remains critical issues in reliable device integration and high-frequency characterization.

Here, we report a concept of a double-side universal soft electronic (double-side USE) platform in which coembedded arrays of polymeric rigid islands and core–shell vias allow site-selective, simultaneous double-side strain-isolating, and vertical-interconnecting functionalities, respectively. All constituents (islands and vias) can be fabricated by single droplet printing of functional inks, and are fully embedded into a common soft substrate with a symmetric square lattice arrangement—a unit structure is 3×3 square lattice comprised of eight islands and one via. Systematic studies on correlation between structural parameters of printed rigid islands (PRIs) and simultaneous double-side strain isolation effect are explored by direct visualization of surface strain field based on digital image correlation (DIC) method, offering an optimized scheme for geometric design of double-side USE platforms. Also, investigation on mechanical stability and high-frequency (>1 MHz) electrical response of stretchable core–shell vias supports practical use in multilayer, stretchable high-speed circuits. In conjunction with the previously reported printing-based electronic functionalization process,[32] the universal concept arrangement of constituents and unique mechanical/electrical feature of the overall platform provide abilities to realize double-side integrated stretchable hybrid electronics with superb circuit design freedom. For a complete example, 36 logic gates interacting with nine vias are integrated into both sides of the soft platform to demonstrate a stretchable 1 MHz binary decoding computational logic circuit. Advanced capabilities of i) forming a fully printable, universal soft platform; ii) implementing multilayer interconnection networks (through several vias) between miniaturized IC devices (≈1.5 mm, one side); and iii) operating high-frequency (≈1 MHz) stretchable computational logic circuits would establish a fundamental framework for stretchable data processing circuit implementation.

2. Result and Discussion

2.1. Concept, Design, and Fabrication of a Double-Side USE Platform

The concept of the double-side USE platform and corresponding functionalities of printing-based implementation of multilayer hybrid electronics are illustrated in **Figure 1**a. Two main features of i) double-side strain isolation with a single island structure and ii) double-side electrical conduction through a core–shell via are concurrently integrated into a common soft substrate (inset image of Figure 1a); establishing a universal basis for system-level, fully printed soft electronics (see Figure 1b for the real image of a double-side integrated stretchable circuit). The fabrication procedure of a double-side USE platform is composed of two major parts: i) rigid island printing (Figure 1cii) and ii) core–shell via formation (Figure 1ciii). All steps were executed by a pneumatic control nozzle-jet printing process; to be specific, single droplets of functional inks printed by a single nozzle were formed into PRIs and stretchable core–shell via structures, respectively. More specific experimental steps were depicted in Figure S1 (Supporting Information). Both components embedded in an elastomer matrix (poly(dimethyl siloxane) (PDMS)) played a role of mechanical and electrical backbones, respectively; leading to a double-side engineered soft substrate for multilayer stretchable hybrid electronics (Figure 1b,civ).

For PRI printing, high concentration poly methylmethacrylate) (PMMA) (M_w 120 000) solution was used. Contrary to the piezoelectric inkjet-printer,[32] a pneumatically controllable nozzle-jet printer was employed to dispense high concentration (4–20 wt%) polymeric ink with high viscosity. Hence, just a single shot printing (air pressure = 150 kPa, dispensing duration time < 4 s) of proper ink completely formed a PRI that possessed a sufficient level of size and robustness in company with a room temperature drying process. For effective double-side strain isolation enabled by single embedded PRI structures, morphology engineering of PRIs was conducted by tuning the concentration of PMMA solution ink (Figure 1d).

For stretchable via fabrication, a single droplet of nickel (Ni) microparticles (average diameter < 5 µm)/PDMS (5:1 mixing ratio)/silicone resin (OE-6630, Dow Corning) mixture ink was printed as a seed structure (Figure 1e, left). After being embedded inside the PDMS matrix (Figure 1e, middle), the printed mixture was exposed to a highly focused magnetic field that was driven by a pair of high aspect ratio (≈9) iron structures and magnets (≈0.17 T) (Figure S2, Supporting Information). Interacting with a focused magnetic field, ferromagnetic Ni microparticles were self-assembled into the center of the mixture; by contrast, the shape of PDMS and silicone resin residues was not affected by the magnetic field, spontaneously constructing a core (Ni/PDMS(5:1)/silicone resin)–shell (PDMS(5:1)/silicone) structure (Figure 1e, right). This vertical alignment of Ni particles offered huge increase in electrical conductivity in vertical direction. Also, the structural peculiarity of the core–shell geometry exhibited modulus-gradient property (Young's modulus: core > shell > elastomer matrix (PDMS(20:1)) for efficient stress absorption; simultaneously satisfying both electrical and mechanical requirements for a stretchable via structure. Moreover, a strong magnetic field attracted Ni particles to the field sources (a pair of iron structures), and therefore embedded Ni particles were moved toward and exposed to both top and bottom surfaces of the platform; consequently forming protruding conductive contacts that facilitated a facile wiring between vias and electronic components. Dimension (thickness and diameter) of vias was optimized with

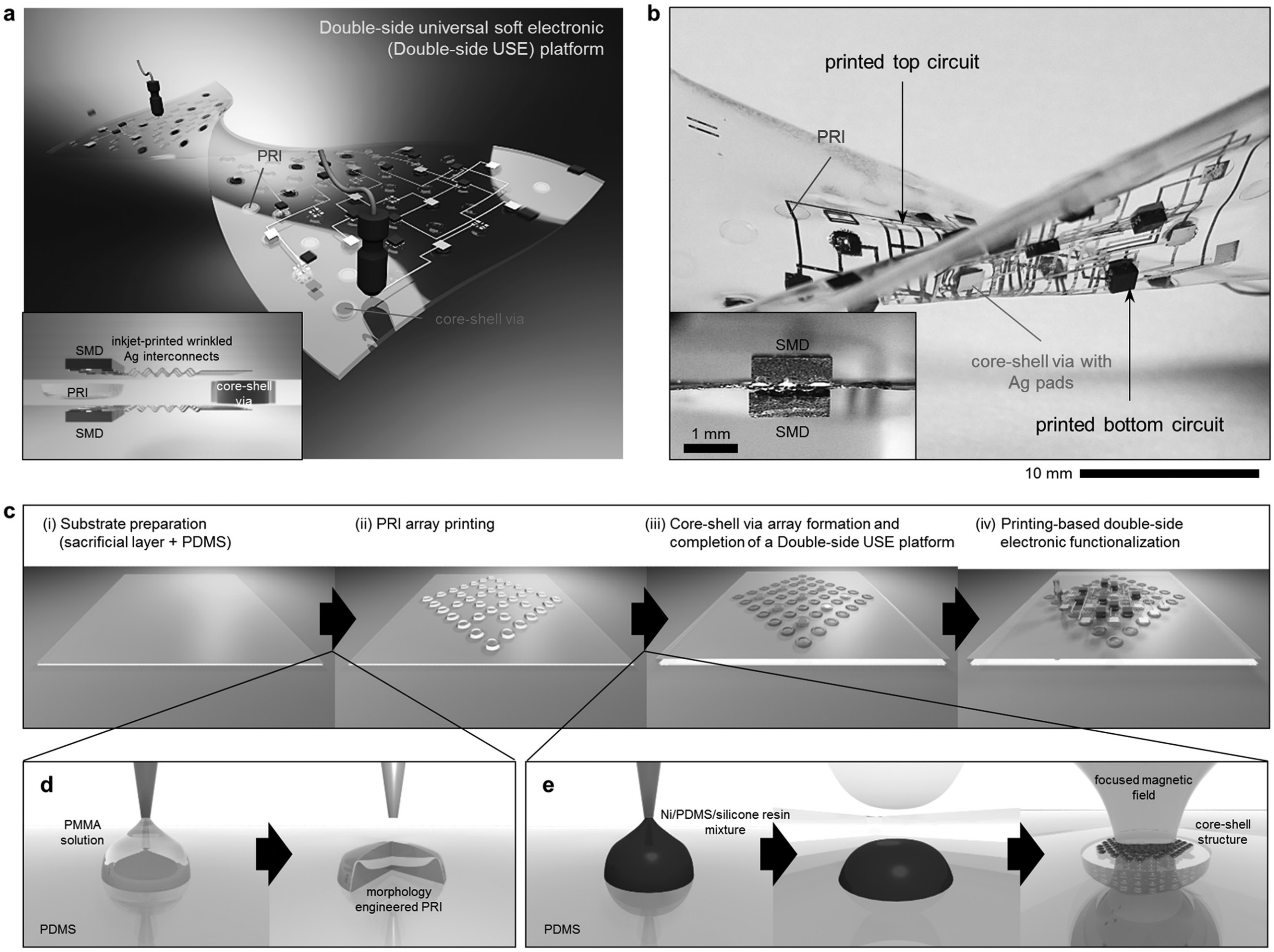

Figure 1. Concept and fabrication of a double-side universal soft electronic (double-side USE) platform. a) Schematic illustration of a double-side USE platform and printing-based system-level integration. Inset illustration shows key features of double-side strain-isolating (or double-side device integration) with a single PRI and double-side vertical-interconnecting with a core–shell via. b) Photograph of a double-side integrated, fully printed, system-level soft electronic circuit fabricated on the double-side USE platform. Inset image: cross-sectional optical image of double-side integrated SMDs. c) Simplified fabrication process: i) substrate preparation, ii) PRI array printing, iii) core–shell via array formation and completion of a double-side USE platform, and iv) printing-based double-side electronic functionalization (detailed fabrication steps of a double-side USE platform are depicted in Figure S1 in the Supporting information). d) Process steps of single droplet printing of a PRI: as-printed PMMA ink (left) is formed into specific island structures (right) along with PMMA concentration. e) Process steps of single droplet printing of a core–shell via: as-printed Ni/PDMS (5:1)/silicone resin mixture (left) is covered by PDMS matrix (middle) and then formed into the core–shell structure (right) driven by a highly focused magnetic field.

form factors that enabled geometric cointegration with PRIs and compact circuit design compared to the previous design (Figure S2, Supporting Information).[33]

In order to involve universality in device integration, PRIs and vias were arranged in a symmetric configuration comprising a 7×7 array whose unit structure was comprised of 8 PRIs and 1 via in a 45°-rotated square lattice configuration (Figure 1c). Considering the specification of the field focusing equipment, the feasible amounts of PRIs and vias in a single platform could be extended to 280 and 81, respectively (Figure S2, Supporting Information). This universal PRI-via arrangement could create a great synergy together with a printing-based electronic functionalization methodology that offered the capability of superb circuit design freedom along with a facile, mask-free fabrication procedure. As a result, printing-based implementation of multilayer stretchable hybrid electronics could be accomplished based on customized networks of inkjet-printed wrinkled silver (Ag) interconnects and multiple type of surface mountable devices (SMDs) without any transfer step.

2.2. Morphology Engineering of PRIs

As one of the main feature of the double-side USE platform, PRIs were fabricated by printing a single droplet of high concentration PMMA ink (M_w 120 000) through a 32G nozzle (**Figure 2**a). A printing condition was stipulated by two parametric values of air pressure and dispensing duration time according to the specific solution concentration. During a room temperature drying process, as-printed PRIs were formed into coffee-ring like island structures. Noting that a coffee-ring structure of a polymeric solution droplet is affected by solution concentration and thus controllable in a stepwise manner,[34,35] we controlled the PRI morphology with diverse PMMA

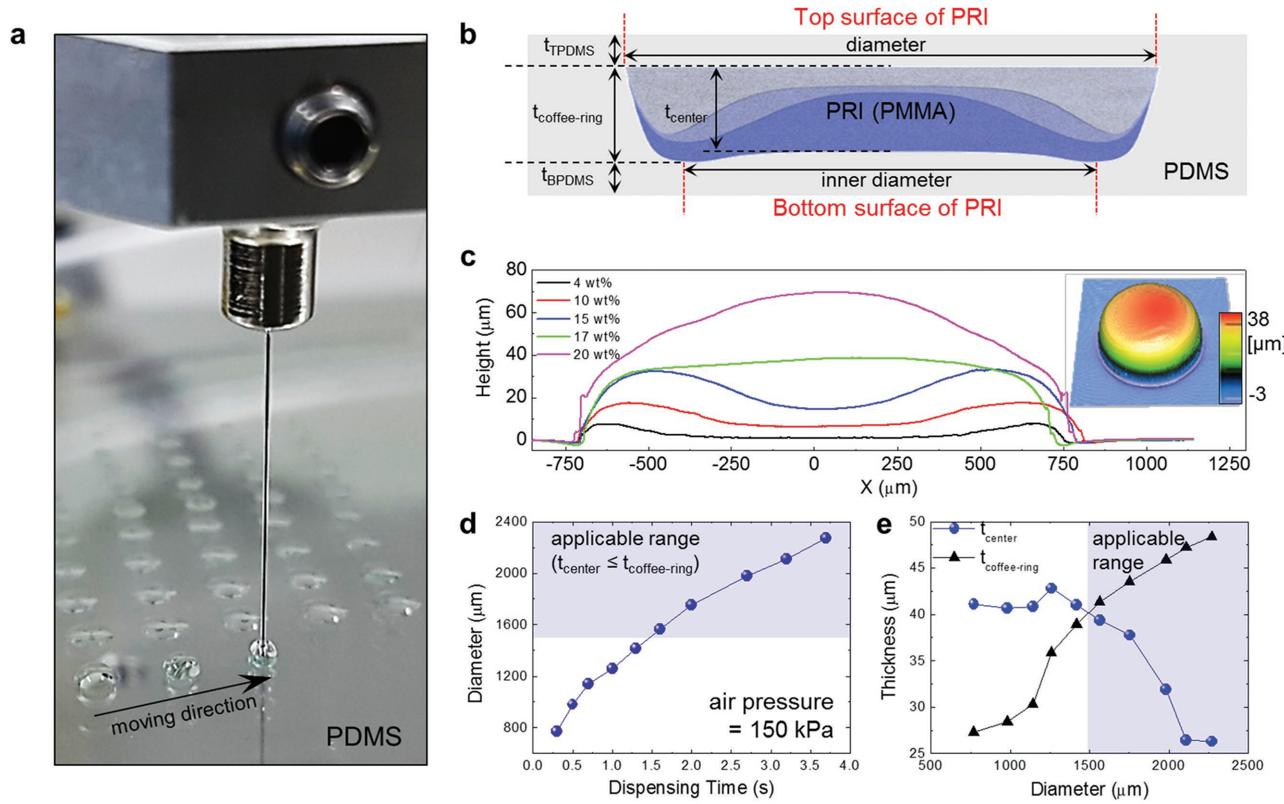

Figure 2. Single droplet printing-based PRI morphology engineering. a) Photograph of real-time single droplet printing of PMMA ink. b) Geometric parameters of a PRI-embedded double-side USE platform. c) Morphology profiles of PRIs with different PMMA concentration (4, 10, 15, 17, 20 wt%). Diameters of each PRI are fixed to ≈1500 μm for an effective parametric sweep. Inset image: 3D surface profile image of a 17 wt% PRI exhibiting a truncated cone structure. Structural engineering of 17 wt% PRIs through varying dispensing duration time with a specific air pressure (150 kPa): diameter (d) and thicknesses (e). Applicable diameter range of 1500–2300 μm is set to satisfy the necessary condition ($t_{center} \le t_{coffee-ring}$) for optimized double-side strain isolation.

concentration (4, 10, 15, 17, 20 wt%). For quantitative analysis, the PRI morphology was investigated in terms of structural parameters: top PDMS thickness (t_{TPDMS}), bottom PDMS thickness (t_{BPDMS}), outer diameter (simply diameter), inner diameter (defined as the distance between two local maxima, mainly coffee-ring structures), thickness at the PRI center (t_{center}), and coffee-ring structure ($t_{coffee-ring}$) (Figure 2b).

Concentration variation in PMMA ink primarily contributed to stepwise splitting of PRI structural parameters at the equal drying condition (Figure 2c) (3D profiler images in Figure S3a in the Supporting Information). For an effective parametric sweep, the diameter was fixed to ≈1500 μm, which was chosen to provide the minimum applicable size to safely preserve a miniaturized SMD light-emitting diode (LED) (1 × 0.6 mm^2) (Figure S4, Supporting Information). Specific printing conditions were adjusted in accordance with ink concentration (4–20 wt%): air pressure level was controlled from 10 to 400 kPa relying on the concentration level to form PRIs of equal size (≈1500 μm); and dispensing duration time was <2 s. Although various pairs of air pressure level and dispensing duration time allowed an analogous PRI morphology (Figure S3b, Supporting Information), air pressure level was regulated by proper duration time (≈2 s). Along with the increase in PMMA concentration, an internal volume—an empty space of a coffee-ring structure—of each PRI was gradually filled with an additional PMMA, and thus the increasing rate of t_{center} was larger than that of $t_{coffee-ring}$ with the fixed diameter that was defined by the effect of contact line pinning. As a result, the PRI morphology was controllable from a coffee-ring structure (<17 wt%) to a truncated cone (≈17 wt%) and a blunt cone structure (>17wt%).

Addressing the balanced structure ($t_{center} \approx t_{coffee-ring}$) at the minimum applicable diameter (≈1500 μm) (Figure 2c) and necessary condition for optimized double-side strain isolation ($t_{center} \le t_{coffee-ring}$) in an entire applicable diameter range (≥1500 μm) (Figure 2d,e), we adopted a 17 wt% PRI as an optimized building block of a double-side USE platform (an in-depth study will be discussed in the next section). A single droplet printing process offered an advantage of controlling a 17 w% PRI diameter only by varying dispensing duration time with a specific air pressure (150 kPa) (Figure 2d). Along with the increase in a diameter, structural parameters of t_{center} and $t_{coffee-ring}$ showed an inverse proportion relationship such that the PRI morphology transformed from the balanced structure ($t_{center} \approx t_{coffee-ring}$) at the minimum applicable diameter (≈1500 μm) to coffee-ring structures ($t_{center} < t_{coffee-ring}$) (Figure 2e). Although t_{center} was gradually reduced, it was kept in a range of 25–40 μm; thereby allowing an adequate robustness in PRIs with a diameter range of 1500–2300 μm, the range of which was delimited to allow compact strain-free areas for standardized SMD components (1.5 × 1.5 mm^2). This transition between deposit types was not a unique morphological feature in 17 wt% PRIs, but a prominent phenomenon that was also observable in PRIs fabricated by different PMMA concentrations (Figure S3c,d, Supporting Information). The only difference is the minimum applicable diameter satisfying the relation, $t_{center} \leq t_{coffee\text{-}ring}$: as the concentration increased, the minimum applicable diameter increased together.[36,37] Based on this principle, a targeted PRI thickness and its applicable diameter range could be employed for desired system implementation.

2.3. Mechanical Stability and Surface Strain Distribution of Engineered PRIs

Figure 3 shows quantitative studies on robustness and strain isolation of morphology engineered PRIs and corresponding double-side USE platforms through DIC-based surface strain mapping over both top and bottom surfaces (see Figure S5 in the Supporting Information for DIC process). Eight distinct samples for each PRI morphology (four for top and four for bottom analysis) were prepared for analysis. All samples were

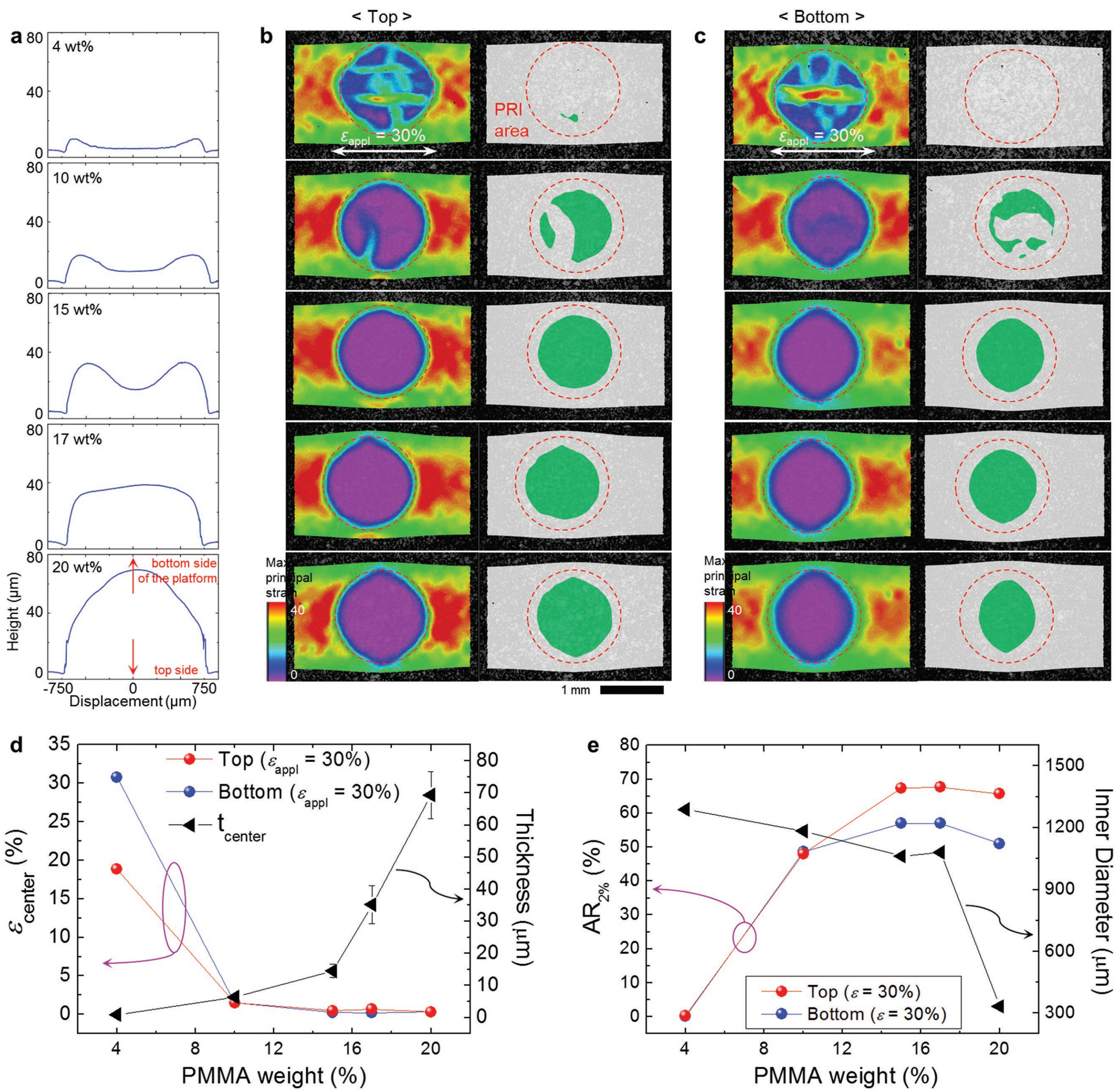

Figure 3. Mechanical stability and surface strain distribution of engineered PRIs. a) Representative morphology profiles of engineered PRIs (PMMA concentration: 4, 10, 15, 17, 20 wt%) with a rescaled form from Figure 2c. Addressing the geometry of a double-side USE platform, top and bottom side of PRIs indicates bottom and top side of double-side USE platforms, respectively. Surface strain distribution (rainbow spectrum) and evaluated strain-free area (green area contrary to white background) on top (b) and bottom (c) surfaces of double-side USE platforms under 30% uniaxial tensile strain. All evaluated strain values were calculated in a form of maximum principal strain. d) Correlation graph between ε_{center} (top and bottom surfaces), t_{center}, and PMMA wt%. e) Correlation graph between $AR_{2\%}$ (top and bottom surfaces), inner diameter, and PMMA wt%.

designed to satisfy the following structural features: PRI diameter ≈1500 μm and t_{TPDMS} = t_{BPDMS} = 20 μm for maximizing strain-free areas without structural delamination (Figure S6, Supporting Information).[32] In particular, surface strain distribution of the whole PRI area (rainbow spectrum) and evaluated strain-free area (green area contrary to white background)—a specific domain with internal strain <2%[33] —on both surfaces were separately explored (top surface: Figure 3b; bottom surface: Figure 3c).

The DIC results suggested that t_{center} played a crucial role in determining robustness of PRIs and double-side USE platforms. Indeed, an insufficient t_{center} (<10 μm in 4, 10 wt% PRIs) gave rise to fatal fracture of the PRI, leading to total destruction of the whole platform. By contrast, PRIs with a sufficient level of t_{center} ($\gtrsim$15 μm in 15, 17, 20 wt% PRIs) perfectly absorbed the applied stress, exhibiting well-defined strain isolation upon the both top and bottom surfaces regardless of structural morphologies (see Figure 3b,c). Focusing on the strain-free areas of robust PRIs, it is noted that difference in the extent of strain-free areas between top and bottom surface inevitably occurred in every samples, regulated by the outer and inner diameter of each embedded PRI (Figure S7, Supporting Information). Taken together, transformation of the PRI morphology from a coffee-ring structure (15 wt%) to a blunt cone structure (20 wt%) dynamically increased t_{center}, and accordingly decreased the inner diameter. As a result, the unexpected reduction of strain-free areas at the bottom surface occurred (Figure 3c).

Additional quantitative studies were made to specify the robustness of the platform in a numerical form and also to figure out the optimized platform based on two kinds of indices: 1) maximum principal strain at the PRI center position (ε_{center}) and 2) strain-free area ratio in PRI ($AR_{2\%}$) that is defined as the ratio of strain-free area (internal strain <2%) to the whole PRI area. Typically, ε_{center} and $AR_{2\%}$ express the PRI robustness and the extent of applicable PRI areas for reliable device protection, respectively. Commensurate with the tendency described in Figure 3b,c, PRIs with a sufficient level of t_{center} (15, 17, 20 wt% PRIs) not only kept their structural morphology without fracture but showed a very low level of ε_{center} (<0.7%) both on the top and bottom surfaces compared to the relatively thin PRIs formed by low concentration ink (Figure 3d). The $AR_{2\%}$ index also supported the foregoing result that the inner diameter of robust PRIs primarily affected the strain-free area of the bottom surface (Figure 3e). The reduced inner diameter of the blunt cone PRI morphology (20 wt%) led to substantial degradation in $AR_{2\%}$ of the bottom surface. On the other hand, despite the difference in PRI morphology, $AR_{2\%}$ of 15 and 17 wt% PRI was kept in a similar level both on the top (≈70%) and bottom surface (≈60%) since both of them possessed comparable structural parameters of outer/inner diameters and t_{PRI}. In this regard, the specific relation of $t_{center} \leq t_{coffee\text{-}ring}$ was defined to be necessary condition for optimized double-side strain isolation. Accordingly, adoption of a 17 wt% PRI as an optimized building block for double-side USE platforms was advocated in terms of robustness and $AR_{2\%}$ as well as structural balance, shortly discussed in Section 2.2. Individual properties of such PRIs suggested that adequate stability (ε_{center} < 1%) and a sufficient extent of strain-free areas ($AR_{2\%}$ ≈ 53% or ≈73% in a length scale) could be guaranteed even under 40% external tensile strain (Figure S8, Supporting Information).

2.4. Structure and Electrical/Mechanical Characteristics of a Stretchable Core–Shell Via

For electrical conduction between both soft surfaces of a double-side USE platform, we previously adopted the solid-phase core–shell structure that was suggested as a promising candidate for a stretchable via.[33] Together with simple fabrication process, this structure showed several advantages in terms of mechanical robustness and good electrical conductivity (≈10 S m^{-1}) based on its structural peculiarity~~-ve~~rtically aligned, modulus gradient geometry. However, the discontinuity in elastic modulus between the core (Ni), shell (silicone resin, OE6630), and soft matrix (20:1 PDMS) often caused unexpected delamination between Ni particles and soft matrix, thus geometric failure of the entire system under a small biaxial stretching condition. Furthermore, due to lack of contemplation in geometric compatibility with coassembled PRIs (or limited consideration in device integration), dimension of core–shell vias was improperly defined. For example, thickness (>300 μm) was too large for implementing a strain-isolated stretchable system.

Improved from our previous approach, this work offered modified geometry and compositions of core–shell vias together with an upgraded magnetic field modulator (Figure S2, Supporting Information) for 2D stretchability (≈30%) and geometric compatibility with PRIs. In addition, a series of electrical characterization was conducted to verify the usefulness of such vias in high-speed demonstration. To be specific, middle-range modulus PDMS (5:1 mixing ratio) was added into the mixture of Ni microparticles and silicone resin, and then the modified Ni/PDMS/silicone resin ink was utilized for the seed of stretchable vias (**Figure 4**a). Through fabrication steps as described in Figure 1e, the clearly defined core–shell structure was obtained with modulus-gradient property (Figure 4b; Figure S9, Supporting Information), concurrently forming vertical conduction paths with densely aligned Ni particles (Figure S10, Supporting Information). Thickness of the via was controlled to be ≈100 μm in ways that offered geometric compatibility with optimized double-side strain-isolating (Figure 4c).

The mechanical and electrical characteristics of the modified core–shell via were investigated by DIC-based surface strain analysis and a series of electrical analyses. Figure 4d shows an average strain level (evaluated from the 50% internal via area) and actual strain distribution of the via under sequential stretching (0–50%). From the result, we figured out that the average strain level of the modified core–shell via (Ni/PDMS/silicone resin-PDMS/silicone resin) (average strain ≈ 3%) slightly increased compared to the previous version (Ni/silicone resin-silicone resin) (average strain ≈ 2%) because the superimposed PDMS (5:1) reduced the effective modulus. Nonetheless, it is notable that the obtained average strain level inside the via was low enough to keep its conductivity under large (ε_{appl} ≈ 100%) and cyclic deformations. To be specific, the range of initial resistance of core–shell vias was around 10–20 Ω, and the resistance versus applied tensile strain profile suggests

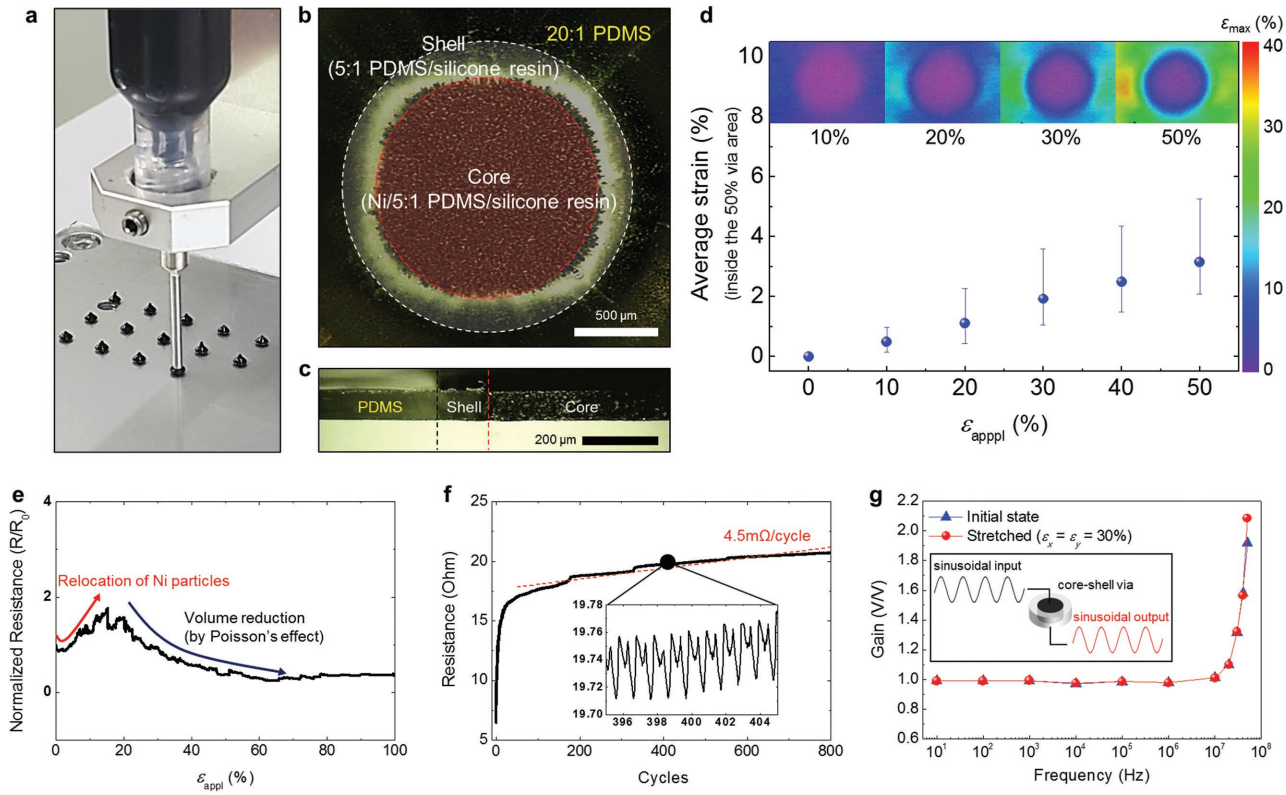

Figure 4. Structure and electrical/mechanical characteristics of a stretchable core–shell via. a) Photograph of real-time single droplet printing of Ni/PDMS (5:1)/silicone mixture ink. b) Optical image of a core–shell via. c) Cross-sectional optical image of a core–shell via. Total thickness of the via was ≈100 µm. d) Surface strain distribution and average strain levels of a core–shell via as a function of ε_{appl} (0–50%). All evaluated strain values were calculated in the form of maximum principal strain. e) Normalized resistance profile curve of a core–shell via during a large deformation (ε_{appl} = 0–100%). Initial resistance of the via was 10–20 Ω, and stretching speed was 1 mm min^{-1}. f) Resistance profile of a core–shell via during repeated deformation (ε_{appl} = 50%). Stretching speed was 100 mm min^{-1}. g) Frequency response of a core–shell via to sinusoidal waves.

that the resistance slightly increased in the low strain regime (ε_{appl} < 20%), started to decrease, and was finally sustained in the high strain regime (ε_{appl} < 100%) (Figure 4e). This is because an elastomer/conductive filler composite with anisotropic filler distribution (like vertical alignment) generally shows negative strain-resistance property, driven by total volume reduction of the composite under uniaxial stretching.[38] For repeated deformation (ε_{appl} = 50%), the via underwent substantial increase in electrical resistance in first few cycles because of the misalignment of off-centered Ni particles that were relatively vulnerable to external stress. However, the resistance was stably maintained under 20 Ω during over 800 cycles with the increasing rate of ≈4.5 mΩ per cycle (Figure 4f). Especially, similar to the resistance profile in Figure 4f, the repeated relocation and volume reduction of the via induced a double-peak profile in every stretching cycle (inset in Figure 4f). Furthermore, it is notable that the addition of buffer PDMS (5:1) greatly contributed to alleviating the localized stress near the core/shell/matrix boundaries so that electrical conductivity of the via was maintained even under cyclic 30% biaxial strain (Figure S11, Supporting Information).

In addition to static electrical analysis, we further explored the frequency response of core–shell vias in terms of input/output gain and signal delay (or phase shift). Figure 4g shows the gain of vias in response to sinusoidal waves with increasing frequency (10 Hz to 50 MHz) both at an initial state and biaxially stretched state (≈30%). The good consistency of gain (≈0.97) in a specific frequency regime (10 Hz to 1 MHz) validated the practical use of core–shell vias in middle-range-speed circuit operation. Also, the sharp increase of gain in the higher frequency regime (>10 MHz) assumed that the resonant frequency (≈(LC)$^{-1/2}$) of the via was located around 100 MHz. Input/output signal delay was also measured by two different input square waves (1 kHz, 1 MHz) (Figure S12, Supporting Information). From the magnified view of signals in Figure S12a (Supporting Information), we identified that the signal delay (mainly related to RC constant) was ≈3.2 ns, and no meaningful difference between delays in 1 kHz and 1 MHz input/output signals (either stretched or not) was observed (Figure S12b, Supporting Information).

2.5. Overall Strain and Conductivity Mapping in a Double-Side USE Platform

A universal concept arrangement of double-side strain-isolating PRIs and double-side vertical-interconnecting stretchable core–shell vias accorded universality in site-selection for device integration and ultimate circuit design (**Figure 5**a). Also, thin (≈100 µm) and soft features of this platform enabled

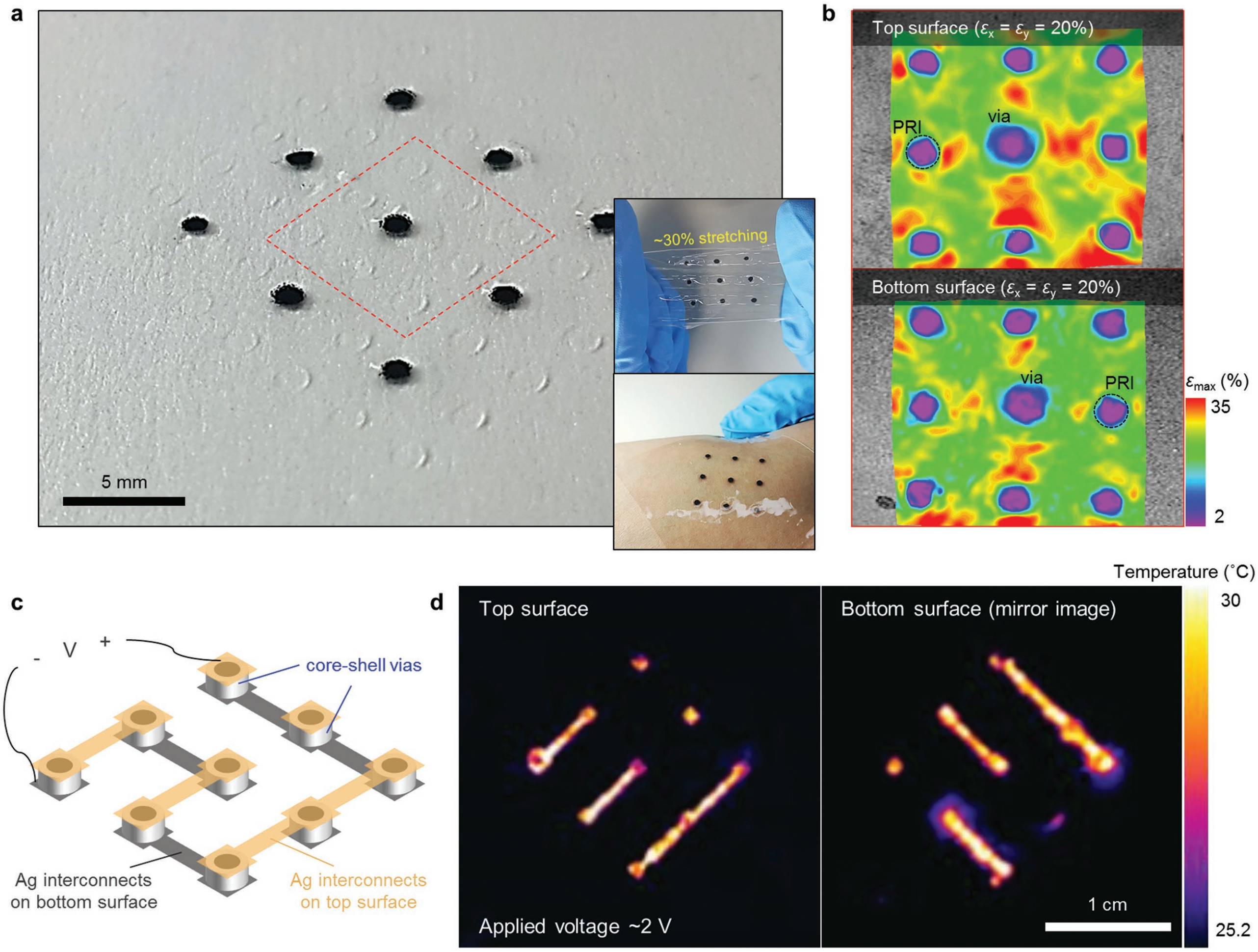

Figure 5. Overall strain and conductivity mapping in a double-side USE platform. a) Photograph of a double-side USE platform comprised of 40 PRIs and 9 core–shell vias. Inset images: optical images of the double-side USE platform under 30% stretching (top) and skin-conforming (bottom). b) Large-area surface strain mapping on both top and bottom surfaces under 20% biaxial strain. c) Schematic illustration of the circuit designed to link nine vias through seven steps of vertical conduction (PRIs are omitted for simplicity). Two-step inkjet-printing processes were carried out for fabrication. d) Temperature gradient mapping of the operating circuit on both top and bottom surfaces. Bias voltage of ≈2 V was applied for a few seconds to obtain observable temperature gradients (≈5 °C). Particularly, for the bottom surface image, a mirror image was utilized for an intuitive observation. These thermographic profiles indirectly visualized the spatial conductivity of double-side interconnected vias and circuit.

maintaining its own properties in response to a series of deformations (insets of Figure 5a). In addition to individual characteristics of the embedded constituents described in Figures 3 and 4, respectively, mechanical and electrical property of overall platform and feasibility of double-side electronic functionalization for stretchable systems needed to be investigated. In this respect, large-area (≈12 × 12 mm^2) surface strain mapping on both surfaces and thermal conductivity mapping for the operating circuit that was simply designed through interconnecting 9 vias were performed to visualize overall characteristics. First, surface strain distribution in a unit domain, containing 3 × 3 components of 8 PRIs and 1 via, of a double-side USE platform was explored in an analogous way as described in previous sections. The difference was that DIC process was carried out under biaxial strains of 0–30% with a reduced resolution in a strain field induced from large-area imaging. The overall DIC result suggested that individual mechanical robustness of PRIs and via was fully reflected that surface strain was exactly isolated both in PRI and via areas, providing arrays of well-defined strain-free areas (Figure 5b). Also, addressing the geometry of PRIs in which the outer diameter (diameter in a top side view) was larger than the inner diameter (diameter in a bottom side view), it is worth noting that the strain localization between the via and adjacent PRIs was more enhanced on the top surface than the bottom surface.

Overall electrical characteristics of the platform was analyzed in terms of reliable vertical interconnection through nine core–shell vias. To visualize the electrical conductivity of such via areas, an indirect way of infrared thermography that mapped thermal gradient of Joule heating generated from operating circuits was adopted. A simple circuit illustrated in Figure 5c (PRIs were omitted for simplicity) was designed to link nine vias through seven steps of vertical conduction, and was fabricated by two-step inkjet-printing processes. Once the bias (≈2 V) was applied to the terminals, printed Ag interconnects and vias started to emit thermal radiation following the equation of $P = V^2/R$; within a few seconds, temperature gradient was observed around the interconnects and vias (Figure 5d). Regulated by biasing time (few seconds) and circuit resistance (≈300–350 Ω), temperature increase was ≈5 °C. The thermographic profiles on the top and bottom surfaces of the double-side USE platform supported the spatial conductivity, and thus double-side electrical conduction path through nine vias.

2.6. Double-Side Integrated Stretchable Binary Decoder

Based on the optimized double-side USE platform, the first demonstration of fully printed, double-side integrated, stretchable computational logic networks that offered an opportunity of "in-system" computing and practical data processing was achieved by means of two-step printing-based electronic functionalization process (see Note and associated Figures S13–S18 in the Supporting Information for step-by-step process flow).[32] The targeted circuit was a kind of decoder that consisted of four inverters, nine multi-input AND gates, and seven multiinput OR gates (Figure S13, Supporting Information), converting high-speed (≈1 MHz) binary code decimal (BCD) to seven-segment signal—the first attempt to operate a soft electronic system in 1 MHz frequency. Addressing the miniaturization of SMDs to reduce bulky characteristics of the whole soft system, we used SMD components whose dimension is standardized to 1.5 × 1.5 mm^2: thus the actual system was comprised of six inverters (two SMDs of three-channel inverter), 12 two-input AND gates (six SMDs of two-channel two-input AND gates), and 18 two-input OR gates (nine SMDs of two-channel two-input OR gates) (Figure S15, Supporting Information). For direct comparison, first, we fabricated such binary decoder in a single-sided layout (Figure S19, Supporting Information). In this layout, the same number of SMDs and equal topology of combinational networks as depicted in Figure S13 (Supporting Information) were employed. For effective double-side integration, on the other hand, the whole circuit layout was divided into two functional groups: top circuit design and bottom circuit design (highlighted in blue and red, respectively, in Figure S13 of the Supporting Information). In the top circuit, combinations of eight inputs (A, A', B, B', C, C', D, D') generated nine output signals (BC', B'C, B'D, BC'D, C'D', CD, B'D', A, BCD') through 12 two-input AND gates. These output signals were connected to nine core–shell vias, and each signal was transmitted to the bottom circuit through vias. In the bottom circuit that is comprised of 18 two-input OR gates, combinational networks between nine signals obtained from nine vias generated individual seven-segment signals (C_0 to C_6) (**Figure 6**a; Figures S13 and S15). Notably, compared to the single-side implementation (Figure S19, Supporting Information), applicable area was increased twice due to the double-side area, whereas the number of utilized PRIs and crossover lines was approximately reduced in half through double-side functionalizing (Figure 6a; **Table 1**).

Figure 6b shows the physical characteristic of the double-side integrated binary decoder that allowed multidimensional deformations such as stretching and twisting. Double-side strain-relief property of PRIs–indirectly advocated by a wrinkle-suppression phenomenon–and inkjet-printed wrinkled Ag interconnects permitted smooth alleviation of applied mechanical stress through an accordion-like physical behavior such that contact areas of integrated SMD chips were safely protected from repetitive elongation (Figure 6c; Figure S19, Supporting Information).[32] Together with the strain-insensitive feature of the core–shell vias (Figure 4e), the complete design of the double-side integrated stretchable decoder showed accurate output characteristics in response to four distinct periodic binary inputs ($A = f_0/8$, $B = f_0/4$, $C = f_0/2$, $A = f_0 = 1.024$ kHz) (Figure 6d). Specifically, the four-bit BCD inputs were decoded into the seven-bit signals that represented descending Arabic numerals. Due to the core functions of the double-side USE platform (double-side strain-isolating and vertical-interconnecting) and reliable inkjet-printed stretchable interconnection networks, the electrical performance of the decoder was stably maintained both at an initial (black line) and a stretched state (red line) (Figure 6d). Furthermore, combinations of reliable high-frequency characteristics of core–shell vias (Figure 4g; Figure S12, Supporting Information), good conductivity (≈2 × 10^6 S m^{-1}) of printed interconnects, and operational reliability of SMD logics enabled 1 MHz operation of the stretchable computational logic circuit (Figure 6e). Although relatively large resistance of printed thin-film (≈500 nm) Ag interconnects (40–50 Ω cm^{-1} with 250 µm average line width) and parasitic capacitance components incurred a perceptible degradation in terms of RC delay and signal distortion, we strongly expected that further studies on improving via characteristics (constituents, geometry, etc.) and reducing resistance and parasitic capacitance (or inductance) of inkjet-printed stretchable interconnects would reach stretchable and fully printable high-frequency (more than MHz) hybrid electronics.

3. Conclusion

In conclusion, we have developed a fully printed, universal soft electronic platform that involved symmetric arrays of site-selective, simultaneous double-side strain-isolating and vertical interconnecting structures for printing-based implementation of multilayer stretchable hybrid electronics. Single droplet printing-based methodology allowed rapid and scalable assembly of both constituents (island and via), and a series of analyses on island morphology, surface strain mapping, and electrical characteristics provided optimized schemes for geometric design of a double-side USE platform. In particular, focusing on high-speed (≈1 MHz) operation of stretchable computing circuits, the proposed concept of double-side USE platforms (especially about characterization of core–shell vias) and printing-based double-side soft integration of hybrid electronic systems would provide a distinct insight into realizing soft, multilayer-integrated, high-speed data-processing systems, and eventually fully integrated system-level soft electronics.

4. Experimental Section

Fabrication of a Fully Printable, Double-Side USE Platform: 5 wt% poly(vinyl alcohol) (Sigma Aldrich, average M_w 31 000–50 000, 87–89%

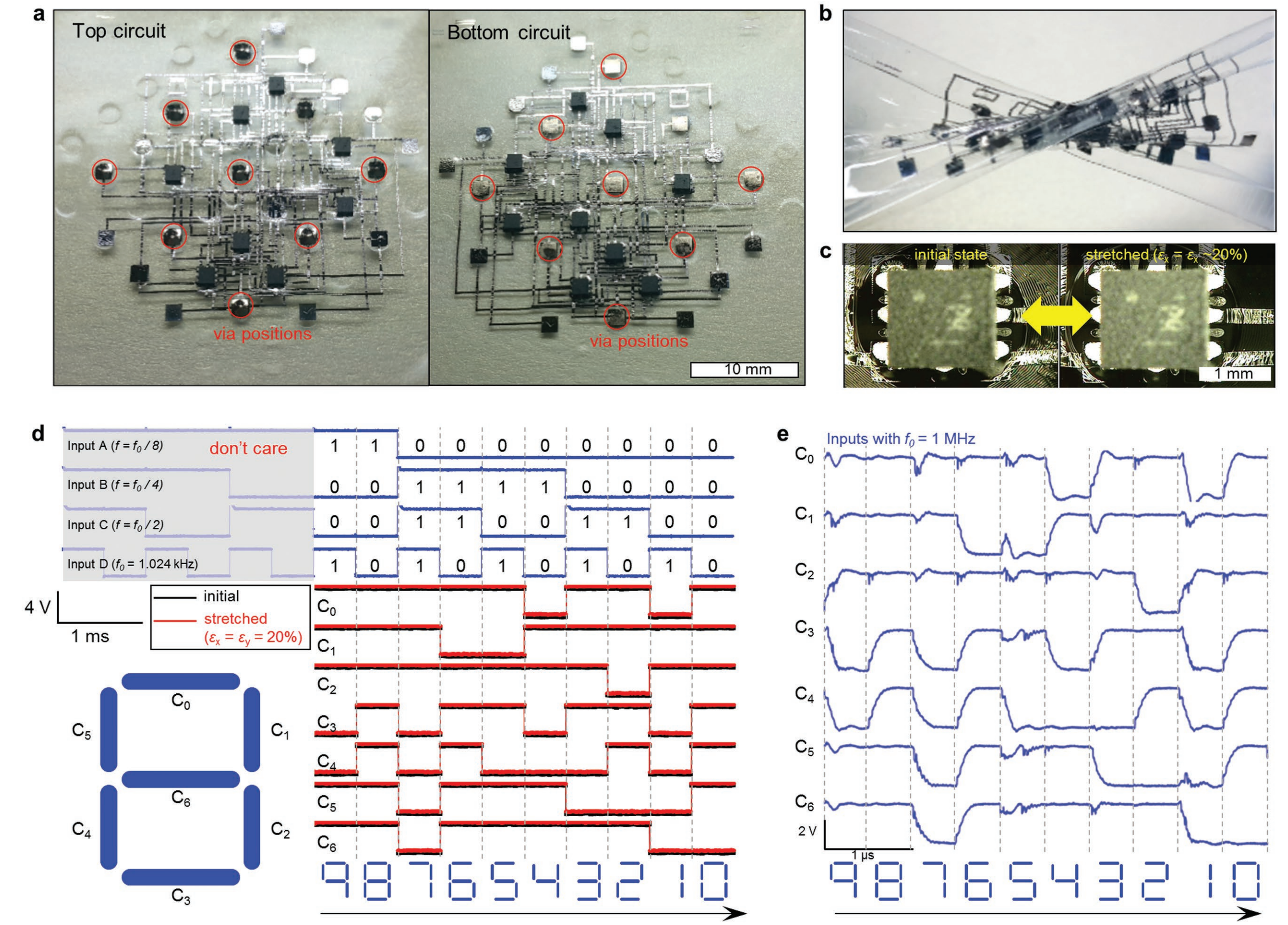

Figure 6. Double-side integrated stretchable binary decoder. a) Photograph of a double-side integrated stretchable BCD-to-seven-segment signal decoder fabricated on the double-side USE platform. The whole circuit was fabricated by two-step printing-based electronic functionalization processes. Through nine vias highlighted in red circles, top and bottom circuits were functionally interconnected. Compared to the single-side implementation in Figure S19 (Supporting Information), utilized PRIs were approximately reduced in half (from 17 to 9), and the number of crossover lines was decreased about 42% (From 197 to 114). b) Photograph of the twisted, double-side integrated binary decoder. c) Optical images of a single SMD device (1.5×1.5 mm^2) supported on a PRI (diameter ≈ 2.3 mm) at an initial state and stretched state (biaxial strain ≈ 20%). Inside the PRI was suppressed the wrinkle formation due to the strain-free effect. d) Input/output characteristics of the decoder in response to the four-bit binary inputs ($f_0/8$, $f_0/4$, $f_0/2$, f_0 = 1.024 kHz). No meaningful degradation of output signals at an initial state (black line) and a stretched state (biaxial strain ≈ 20%, red line) was observed. e) Output characteristics of the decoder operating at f_0 = 1 MHz inputs after 20 times stretching cycles.

hydrolyzed) solution dissolved in deionized (DI) water was spin-coated (1000 rpm, 60 s) upon the cleaned glass substrate (700 µm thickness) and thermally annealed for 1 h at 100 °C for the sacrificial layer. Upon it, a 3×3 array (pitch ≈ 8.48 mm) of thermally releasable tapes (TW-50R150, Tape world) was affixed for predefining via positions. Subsequently, PDMS elastomer (Sylgard 184, Dow Corning), mixed at 20:1 ratio, was poured and spin-coated at 3000 rpm for 60 s to form 20-µm-thick PDMS layer. After 30 min of flattening the PDMS surface, the thin PDMS layer was soft-baked for 30 min at 80 °C. During the soft baking process, thermally releasable tapes were detached from the substrate by foaming-induced loss of adhesion to allow the vias to contact the external circuits. After that, an array of PRIs was fabricated by printing 17 wt% PMMA (Sigma Aldrich, M_w 120 000) solution dissolved in propylene glycol monomethyl ether acetate (Sigma Aldrich, ≥99.5%) through a 32G nozzle (inner diameter ≈ 100 µm) with an air pressure level of 150 kPa for 3 s (Figure 1c(i),d). As-printed PRIs were formed into coffee-ring structures (or truncated-cone structures) with a ≈2.3-mm-diameter during a room temperature drying process, and then heated to remove residual solvents. After PRIs were fully dried, a mixture of Ni microparticles/PDMS(5:1)/silicone resin (OE6630, Dow Corning) with mixing ratio of 5:2:3 (weight ratio) was printed at the predefined positions as a seed structure of a core–shell via. An air pressure level of 300 kPa was applied for 0.2 s to form ≈2 mm diameter seed structures. After an array of seed structures was defined (pitch ≈8.48 mm), PDMS was poured

Table 1. Comparison between single- and double-side integrated stretchable binary decoders.

Circuit type	No. of circuit layers (including crossover circuits)	No. of utilized PRIs	No. of utilized vias	No. of crossover lines
Single-side integrated (Figure S19, Supporting Information)	2	17	0	197
Double-side integrated (Figure 6a)	4	9	9	114

onto the entire substrate, and highly focused magnetic field (magnetic field density at the source magnet ≈0.17 T)—driven by pairs of high aspect ratio iron structures ($1 \times 1 \times 9$ mm^3, aspect ratio ≈ 9) (Figure S2, Supporting Information)—was applied to the both surfaces to assemble Ni microparticles into the center area, resulting in a core–shell via structure. After being fully cured, the whole platform was immersed into DI water to dissolve the sacrificial layer, then the double-side USE platform was finally obtained.

Surface Strain Mapping based on DIC: Upon the PRI (or via) area, baking powders (average particle size ≈20 µm) were dredged to form randomly distributed speckle patterns. In interesting areas, sequence of image frames was captured by an optical microscope as the platform was slowly elongated with a strain level of 0–40%. Each image had a resolution of 720×480 pixels, and pixels in images were used to evaluate the relative movement of each speckle pattern based on an image-processing tool (Vic-2D 2009, Correlated Solutions). The evaluated displacement field data could be converted into Hencky strain field, and therefore the overall interesting areas could be mapped with maximum principal strain (Figure S5, Supporting Information).

Electrical Measurement of Stretchable Core–Shell Vias: For stable electrical measurement under sequential mechanical deformation of the soft substrate, liquid metal (EGaIn, Eutectic gallium indium, Sigma-Aldrich) was used as a contact material. During measurement, a small amount of EGaIn was placed above and below the vias to ensure a stable electrical contact between vias and external wirings for resistance measurement. For uniaxial stretching tests, width and length of the via-embedded substrate was defined to be 5 and 10 mm, respectively, and the substrate was slowly elongated at a speed of 1 mm min^{-1}. For cyclic measurement, the repeated strain (50%) was applied at a speed of 100 mm min^{-1}. While the sample was sequentially elongated, electrical resistance of the via was continuously measured and recorded by Keithley 2420 and Labview program.

Thermal Conductivity Mapping of a Double-Side USE Platform: A sample circuit was prepared by two-step inkjet-printing processes together with a slight biaxial prestrain (≈5%) to avoid undesired crack on printed Ag interconnects (see Figure 5c for specific design of the circuit). Then, bias voltage of ~2V was applied to both terminals through liquid metal (EGaIn) contacts for a few seconds. While the bias was applied, thermographic images were captured by a thermal imaging camera (T420, FLIR). Specific setting conditions were as follow: emissivity = 0.86, reflected temperature = 24 °C, ambient temperature = 24 °C, and humidity = 19%.

Printing-Based Double-Side Electronic Functionalization for a Stretchable 1 MHz Binary Decoder Demonstration: Detailed process for demonstration is available in the Supporting Information.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

core–shell vias, double-side integration, rigid islands, stretchable hybrid electronics, universal platforms

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[1] D.-H. Kim, N. Lu, R. Ma, Y.-S. Kim, R.-H. Kim, S. Wang, J. Wu, S. M. Won, H. Tao, A. Islam, K. J. Yu, T.-I. Kim, R. Chowhury, M. Ying, L. Xu, M. Li, H.-J. Chung, H. Keum, M. McCormick, P. Liu, Y.-W. Zhang, F. G. Omenetto, Y. Huang, T. Coleman, J. A. Rogers, *Science* **2011**, *333*, 838.
[2] C.-C. Kim, H.-H. Lee, K. H. Oh, J.-Y. Sun, *Science* **2016**, *353*, 682.
[3] A. Koh, D. Kang, Y. Xue, S. Lee, R. M. Pielak, J. Kim, T. Hwang, S. Min, A. Banks, P. Bastien, M. C. Manco, L. Wang, K. R. Ammann, K.-I. Jang, P. Won, S. Han, R. Ghaffari, U. Paik, M. J. Slepian, G. Balooch, Y. Huang, J. A. Rogers, *Sci. Transl. Med.* **2016**, *8*, 366ra165.
[4] A. P. Gerratt, H. O. Michaud, S. P. Lacour, *Adv. Funct. Mater.* **2015**, *25*, 2287.
[5] D. J. Lipomi, M. Vosgueritchian, B. C.-K. Tee, S. L. Hellstrom, J. A. Lee, C. H. Fox, Z. Bao, *Nat. Nanotechnol.* **2011**, *6*, 788.
[6] S. Y. Kim, S. Park, H. W. Park, D. H. Park, Y. Jeong, D. H. Kim, *Adv. Mater.* **2015**, *27*, 4178.
[7] D.-H. Kim, N. Lu, Y. Huang, J. A. Rogers, *MRS Bull.* **2012**, *37*, 226.
[8] C. Keplinger, J.-Y. Sun, C. C. Foo, P. Rothemund, G. M. Whitesides, Z. Suo, *Science* **2013**, *341*, 984.
[9] Y. Kim, J. Zhu, B. Yeom, M. D. Prima, X. Su, J.-G. Kim, S. J. Yoo, C. Uher, N. A. Kotov, *Nature* **2013**, *500*, 59.
[10] A. Hirsch, H. O. Michaud, A. P. Gerratt, S. Mulatier, S. P. Lacour, *Adv. Mater.* **2016**.
[11] C. Hou, T. Huang, H. Wang, H. Yu, Q. Zhang, Y. Li, *Sci. Rep.* **2013**, *3*, 3138.
[12] D. Brosteaux, F. Axisa, M. Gonzalez, J. Vanfleteren, *IEEE Electron Dev. Lett.* **2007**, *28*, 552.
[13] Y. Sun, W. M. Choi, H. Jiang, Y. Y. Huang, J. A. Rogers, *Nat. Nanotechnol.* **2006**, *1*, 201.
[14] J. A. Fan, W.-H. Yeo, Y. Su, Y. Hattori, W. Lee, S.-Y. Jung, Y. Zhang, Z. Liu, H. Cheng, L. Falgout, M. Bajema, T. Coleman, D. Gregoire, R. J. Larsen, Y. Huang, J. A. Rogers, *Nat. Commun.* **2014**, *5*, 3266.
[15] T. Cheng, Y. Zhang, W.-Y. Lai, W. Huang, *Adv. Mater.* **2015**, *27*, 3349.
[16] J. W. Boley, E. L. White, G. T.-C. CHiu, R. K. Kramer, *Adv. Funct. Mater.* **2014**, *24*, 3501.
[17] R. Matsuzaki, K. Tabayashi, *Adv. Funct. Mater.* **2015**, *25*, 3806.
[18] J. Ren, Y. Zhang, W. Bai, X. Chen, Z. Zhang, X. Fang, W. Weng, Y. Wang, H. Peng, *Angew. Chem. Int. Ed.* **2014**, *53*, 7864.
[19] Y.-H. Lee, J.-S. Kim, J. Noh, I. Lee, H. J. Kim, S. Choi, J. Seo, S. Jeon, T.-S. Kim, J.-Y. Lee, J. W. Choi, *Nano Lett.* **2013**, *13*, 5753.
[20] A. M. Gaikwad, A. M. Zamarayeva, J. Rousseau, H. Chu, I. Derin, D. A. Steingart, *Adv. Mater.* **2012**, *24*, 5071.
[21] S. I. Park, G. Shin, J. G. McCall, R. Al-Hasani, A. Norris, L. Xia, D. S. Brenner, K. N. Noh, S. Y. Bang, D. L. Bhatti, K.-I. Jang, S.-K. Kang, A. D. Mickle, G. Dussor, T. J. Price, R. W. Gereau, M. R. Bruchas, J. A. Rogers, *Proc. Natl. Acad. Sci. USA* **2016**, *113*, E8169.
[22] J. Kim, A. Banks, H. Cheng, Z. Xie, S. Xu, K.-I. Jang, J. W. Lee, Z. Liu, P. Gutruf, X. Huang, P. Wei, F. Liu, K. Li, M. Dalal, R. Ghaffari, X. Feng, Y. Huang, S. Gupta, U. Paik, J. A. Rogers, *Small* **2015**, *11*, 906.
[23] J. Ge, L. Sun, F.-R. Zhang, Y. Zhang, L.-A. Shi, H.-Y. Zhao, H.-W. Zhu, H.-L. Jiang, S.-H. Yu, *Adv. Mater.* **2016**, *28*, 722.

[24] G. Y. Bae, S. W. Pak, D. Kim, G. Lee, D. H. Kim, Y. Chung, K. Cho, *Adv. Mater.* **2016**, *28*, 5300.
[25] Y. Joo, J. Byun, N. Seong, J. Ha, H. Kim, S. Kim, T. Kim, H. Im, D. Kim, Y. Hong, *Nanoscale* **2015**, *7*, 6208.
[26] J. Liang, L. Li, X. Niu, Z. Yu, Q. Pei, *Nat. Photonics* **2013**, *7*, 817.
[27] J. Wang, C. Yan, G. Cai, M. Cui, A. L.-S. Eh, P. S. Lee, *Adv. Mater.* **2016**, *28*, 4490.
[28] W. Gao, S. Emaminejad, H. Y. Y. Nyein, S. Challa, K. Chen, A. Peck, H. M. Fahad, H. Ota, H. Shiraki, D. Kiriya, D.-H. Lien, G. A. Brooks, R. W. Davis, A. Javey, *Nature* **2016**, *529*, 509.
[29] S. Xu, Y. Zhang, L. Jia, K. E. Mathewson, K.-I. Jang, J. Km, H. Fu, X. Huang, P. Chava, R. Wang, S. Bhole, L. Wang, Y. J. Na, Y. Guan, M. Flavin, Z. Han, Y. Huang, J. A. Rogers, *Science* **2014**, *344*, 70.
[30] Y. Liu, J. J. S. Norton, R. Qazi, Z. Zou, K. R. Ammann, H. Liu, L. Yan, P. L. Tran, K.-I. Jang, J. W. Lee, D. Zhang, K. A. Kilian, S. H. Jung, T. Bretl, J. Xiao, M. J. Slepian, Y. Huang, J.-W. Jeong, J. A. Rogers, *Sci. Adv.* **2016**, *2*, e1601185.
[31] Y. Khan, M. Garg, Q. Gui, M. Schadt, A. Gaikwad, D. Han, N. A. D. Yamamoto, P. Hart, R. Welte, W. Wilson, S. Czarnecki, M. Poliks, Z. Jin, K. Ghose, F. Egitto, J. Turner, A. C. Arias, *Adv. Funct. Mater.* **2016**, *26*, 8764.
[32] J. Byun, B. Lee, E. Oh, H. Kim, S. Kim, S. Lee, Y. Hong, *Sci. Rep.* **2017**, *7*, 45328.
[33] E. Oh, J. Byun, B. Lee, S. Kim, D. Kim, J. Yoon, Y. Hong, *Adv. Electron. Mater.* **2017**, *3*, 1600517.
[34] T. Okuzono, M. Kobayashi, M. Doi, *Phys. Rev. E* **2009**, *80*, 021603.
[35] K. Ozawa, E. Nishitani, M. Doi, *Jpn. J. Appl. Phys.* **2005**, *44*, 4229.
[36] H. M. Gorr, J. M. Zueger, J. A. Barnard, *J. Phys. Chem. B* **2012**, *116*, 12213.
[37] K. L. Maki, S. Kumar, *Langmuir* **2011**, *27*, 11347.
[38] S. Kim, J. Byun, S. Choi, D. Kim, T. Kim, S. Chung, Y. Hong, *Adv. Mater.* **2014**, *26*, 3094.

